ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 1: ShopSmart Retail Analytics & AI Implementation Project

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 1
Project	ShopSmart Retail Analytics & AI Implementation
Project Title	Delivering a Retail Analytics and Artificial Intelligence Solution for ShopSmart Retail Group
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 1 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 1 Submission Form

1. DATA ANALYTICS TRACK

Project Title

Mission 01: Retail Sales Analysis with Microsoft Excel

Business Scenario

Client

ShopSmart Retail Ltd

ShopSmart Retail Ltd. operates multiple retail stores across Nigeria. The company sells thousands of products across different categories and receives orders from thousands of customers every month.

Management has noticed fluctuations in sales, inventory shortages, and increasing product returns. They have hired **AnalystLab Africa** to analyze the business data and provide actionable recommendations.

You have been assigned as a **Junior Data Analyst**.

Your task is to clean the data, analyze it using Microsoft Excel, create dashboards, and present business insights.

Learning Objectives

By the end of Week 1, interns should be able to:

- Import CSV files into Excel
- Format data professionally
- Clean messy data
- Create Excel Tables
- Use Excel formulas
- Use XLOOKUP/VLOOKUP
- Build Pivot Tables
- Create Pivot Charts
- Build an Executive Dashboard
- Present business insights

Dataset Provided

Interns receive:

Customers.csv
Products.csv
Orders.csv
Stores.csv
Returns.csv
Employees.csv
Suppliers.csv
Inventory.csv
Order_Items.csv
Data_Dictionary.xlsx

PART 1 — Data Understanding (10 Marks)

Open all datasets.

Answer:

1. How many datasets were provided?
2. What does each dataset represent?
3. Identify the Primary Key of each table.
4. Identify possible Foreign Keys.
5. Draw the relationship between the tables.

Deliverable:

A one-page explanation.

PART 2 — Data Cleaning (15 Marks)

Using Excel,

Perform the following:

- Remove duplicate rows
- Check missing values
- Correct inconsistent date formats
- Ensure currency fields are numeric
- Convert datasets into Excel Tables-
- Rename worksheets appropriately

Deliverable:

Cleaned Excel Workbook.

PART 3 — Excel Formulas (20 Marks)

Use formulas to answer the following.

1. Total Revenue
2. Total Quantity Sold
3. Average Selling Price
4. Average Order Value
5. Highest Sales Amount
6. Lowest Sales Amount
7. Number of Orders
8. Number of Customers
9. Number of Products
10. Number of Stores

PART 4 — Lookup Functions (15 Marks)

Using XLOOKUP (or VLOOKUP),

Retrieve:

Customer Name

Store Name

Product Name

Category

Brand

Selling Price

Supplier Name

PART 5 — Pivot Tables (20 Marks)

Create Pivot Tables showing:

Sales by Month

Sales by Store

Sales by Category

Sales by Brand

Sales by State

Sales by Customer Gender

Sales by Membership Level

Top 10 Products

Bottom 10 Products

Top 10 Customers

PART 6 — Charts (10 Marks)

Create the following charts.

Monthly Sales Trend

Revenue by Category

Revenue by Store

Top Products

Sales by Region

Return Rate by Category

PART 7 — Dashboard (30 Marks)

Design an Executive Dashboard containing:

KPI Cards

Total Revenue

Total Orders

Total Customers

Average Order Value

Return Rate

Inventory Value

Charts

Monthly Sales Trend

Revenue by Category

Revenue by Store

Top Products

Sales by State

Interactive Filters

Year

Month

Region

Store

Category

Membership Level

PART 8 — Business Questions (30 Marks)

Answer the following.

1. Which product generated the highest revenue?
2. Which store performed best?
3. Which category sells the most?
4. Which month generated the highest revenue?
5. Which customers spend the most?
6. Which products are frequently returned?
7. Which supplier supplies the highest number of products?
8. Which stores have low inventory?
9. Which region performs best?
10. Which product category should management invest in?

PART 9 — Business Recommendations (20 Marks)

Write a one-page report.

Include:

Three key findings.

Three business recommendations.

One risk identified.

One opportunity identified.

Final Deliverables

Each intern submits:

```
Week1_FirstName_LastName/
├── ShopSmart_Analysis.xlsx
```

```
├── Executive_Dashboard.pdf
├── Business_Report.pdf
└── README.md
```

PART 10 — ASSESSMENT CRITERIA

Section	Marks
Data Understanding	10
Data Cleaning	15
Excel Formulas	20
Lookup Functions	15
Pivot Tables	20
Charts	10
Dashboard	30
Business Questions	30
Business Report	20
Bonus (Creativity & Presentation)	**10**
Total	**180 Marks**

2. DATA SCIENCE TRACK

Week 1 – Introduction to Data Science with Python

Mission DS01: Exploratory Data Analysis (EDA) on Retail Sales Data

Client

ShopSmart Retail Ltd.

ShopSmart Retail Ltd. has accumulated years of customer, product, sales, inventory, and returns data. Before building predictive models, management wants to understand the data and identify hidden patterns.

As a Junior Data Scientist, your responsibility is to perform an Exploratory Data Analysis (EDA) and present insights that will guide future machine learning projects.

Learning Objectives

By the end of Week 1, interns should be able to:

- Understand the Data Science lifecycle
- Load datasets into Python
- Clean data
- Explore data